

# Day 3 – Module 6: Mixed References

There are 2 types:

## Type 1: Fixed Column

=\$A1

- Column A is fixed.
- Row changes when dragged.

Examples:

=\$A1

=\$A2

=\$A3

|           | Jan | Feb | Mar |           |     |     |
|-----------|-----|-----|-----|-----------|-----|-----|
| Product A | 100 | 120 | 150 | Product A | Jan | Feb |
| Product B | 200 | 220 | 250 | Product B | Jan | Feb |
| Product C | 300 | 320 | 350 | Product C | Jan | Feb |

| Formula | Meaning            |
|---------|--------------------|
| A1      | Relative Reference |
| \$A\$1  | Absolute Reference |
| \$A1    | Fixed Column       |
| A\$1    | Fixed Row          |

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Your screenshot proves both mixed references are working correctly.

## **\$A2 (Fixed Column)**

Results:

- Product A
- Product B
- Product C

Column **A** stayed fixed, while rows changed.

## **B\$1 (Fixed Row)**

Results:

- Jan
- Feb

Row **1** stayed fixed, while columns changed from **B** → **C**.

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# **Day 3 Completed**

## **Topics Covered**

- ✓ Number Formatting
- ✓ Date Formatting
- ✓ Percentage Formatting
- ✓ Relative Reference (**A1**)
- ✓ Absolute Reference (**\$A\$1**)
- ✓ Mixed Reference (**\$A1**, **A\$1**)
- ✓ Formula Dragging

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| Reference Type       | Example | Behavior                  |
|----------------------|---------|---------------------------|
| Relative             | A1      | Row & Column change       |
| Absolute             | \$A\$1  | Nothing changes           |
| Mixed (Column Fixed) | \$A1    | Column fixed, Row changes |
| Mixed (Row Fixed)    | A\$1    | Row fixed, Column changes |

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## 1. Relative Reference (A1)

Formula:

=A1

If you drag it **one column right**:

=B1

If you drag it **one row down**:

=A2

### Rule

- ✓ Row changes
- ✓ Column changes

▼ | fx =A1

| A | B   | C   | D   | E | F | G |
|---|-----|-----|-----|---|---|---|
| 1 | 100 | 200 | 300 | 1 |   |   |
| 2 | 400 | 500 | 600 |   |   |   |
| 3 | 700 | 800 | 900 |   |   |   |
|   |     |     |     |   |   |   |
|   |     |     |     |   |   |   |

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▼ | fx =A1

| A | B   | C   | D   | E | F   | G |  |
|---|-----|-----|-----|---|-----|---|--|
| 1 | 100 | 200 | 300 | 1 | 100 |   |  |
| 2 | 400 | 500 | 600 | 2 |     |   |  |
| 3 | 700 | 800 | 900 |   |     |   |  |
|   |     |     |     |   |     |   |  |
|   |     |     |     |   |     |   |  |

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## 2. Absolute Reference (**\$A\$1**)

Formula:

=A\$1

Drag anywhere:

=A\$1

still

=A\$1

fx =\$a\$1

| A | B   | C   | D   | E     | F | G |
|---|-----|-----|-----|-------|---|---|
| 1 | 100 | 200 | 300 | =a\$1 |   |   |
| 2 | 400 | 500 | 600 |       |   |   |
| 3 | 700 | 800 | 900 |       |   |   |
|   |     |     |     |       |   |   |
|   |     |     |     |       |   |   |

3    fx =\$A\$1

| A | B   | C   | D   | E | F | G |
|---|-----|-----|-----|---|---|---|
| 1 | 100 | 200 | 300 | 1 |   |   |
| 2 | 400 | 500 | 600 | 1 |   |   |
| 3 | 700 | 800 | 900 | 1 |   |   |
|   |     |     |     |   |   |   |
|   |     |     |     |   |   |   |

## Rule

- ✗ Row doesn't change
- ✗ Column doesn't change

Think:

"Lock everything"

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### 3. Mixed Reference – Fixed Column (\$A1)

Formula:

=\$A1

Drag **down**:

=\$A2

=\$A3

Drag **right**:

=\$A1

still Column A.

▼ | fx =\$A1

| B   | C   | D   | E | F | G | H |
|-----|-----|-----|---|---|---|---|
| 100 | 200 | 300 | 1 | 1 | 1 |   |
| 400 | 500 | 600 | 2 |   |   |   |
| 700 | 800 | 900 | 3 |   |   |   |
|     |     |     |   |   |   |   |
|     |     |     |   |   |   |   |

#### Rule

✓ Row changes

✗ Column doesn't change

Think:

"Column A is locked"

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## 4. Mixed Reference – Fixed Row (A\$1)

Formula:

=A\$1

Drag **right**:

=B\$1

=C\$1

=D\$1

Drag **down**:

=A\$1

Row 1 stays fixed.

|    |          |     |     |   |   |     |     |     |   |  |
|----|----------|-----|-----|---|---|-----|-----|-----|---|--|
| F1 | fx =A\$1 |     |     |   |   |     |     |     |   |  |
|    | B        | C   | D   | E | F | G   | H   | I   | J |  |
| 1  | 100      | 200 | 300 | 1 | 1 | 100 | 200 | 300 |   |  |
| 2  | 400      | 500 | 600 | 2 | 1 |     |     |     |   |  |
| 3  | 700      | 800 | 900 | 3 | 1 |     |     |     |   |  |
| 4  |          |     |     |   |   |     |     |     |   |  |
| 5  |          |     |     |   |   |     |     |     |   |  |
| 6  |          |     |     |   |   |     |     |     |   |  |

## Rule

✗ Row doesn't change

✓ Column changes

Think:

"Row 1 is locked"

Example:

| Formula | Result |
|---------|--------|
|---------|--------|

|      |    |
|------|----|
| A\$1 | A1 |
|------|----|

|      |    |
|------|----|
| B\$1 | B1 |
|------|----|

|      |    |
|------|----|
| C\$1 | C1 |
|------|----|

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## Easy Memory Trick

The \$ locks whatever comes immediately after it.

**Dollar before Column**



\$A1

Locks:

Column A

**Dollar before Row**

A\$1

Locks:

Row 1

**Dollar before Both**

\$A\$1

Locks:

Column A and Row 1

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## One-Line Summary

| Formula | What is Locked? |
|---------|-----------------|
| A1      | Nothing         |
| \$A1    | Column          |
| A\$1    | Row             |
| \$A\$1  | Row + Column    |